Consistency of Compressible Momentum Thickness Formula in Schlichting's Bounadry-Layer Theory

A short proof concerning Schlichting's Eqs. (10.92) and (10.96)

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1 Summary

This report concerns H. Schlichting and K. Gersten's *Boundary-Layer Theory*, 9th ed. (Springer, 2017), specifically Eqs. (10.92) and (10.96) [1]. The momentum thickness printed as Eq. (10.96), denoted by θ_S below, is

$$\theta_\mathrm{S} = \int_0^h \frac{\rho u}{\rho_e u_e}\left(1 - \frac{\rho u}{\rho_e u_e}\right) \mathrm{d}y.$$

When read literally, this expression is not generally compatible with the momentum-integral equation printed as Eq. (10.92) for a variable-density profile. With the standard zero-defect normalization, compatibility instead requires

$$\delta_2 = \theta_\mathrm{c} = \int_0^h \frac{\rho u}{\rho_e u_e}\left(1 - \frac{u}{u_e}\right) \mathrm{d}y.$$

Importantly, the same corrected definition appears in similar foundational books by White and Majdalani, Eq. (7-75), and Anderson, Eq. (6.137) [2, 3]; Section 3.3 discusses this independent cross-check.

- **Inconsistency localized:** the second density factor in the deficit term of the printed Eq. (10.96) changes the transported momentum flux from ρu^2 to $\rho^2 u^2/\rho_e$.
- **Compatible definition:** the given momentum-integral equation Eq. (10.92) requires the conventional velocity-deficit form of the momentum thickness, shown above, once the common zero-defect normalization is imposed.

2 Consistency proof

Scope

Schlichting's momentum-integral equation, Eq. (10.92), is accepted as given, including the assumptions under which it was derived. The purpose is only to determine which profile integral can represent the momentum thickness in that equation. A fixed upper limit h is taken in uniform edge flow; it may equally be replaced by the boundary-layer edge or by infinity.

Result. With the standard zero-defect normalization, the momentum thickness compatible with Schlichting's Eq. (10.92) is

$$\delta_2 = \theta_\mathrm{c} = \int_0^h \frac{\rho u}{\rho_e u_e}\left(1 - \frac{u}{u_e}\right) \mathrm{d}y. \tag{1}$$

The expression printed as Schlichting's Eq. (10.96), if read literally, is not generally compatible with Eq. (10.92) unless an additional density-defect term is retained.

Proof. Schlichting's momentum-integral equation Eq. (10.92) is

$$\delta_2' + \frac{\delta_2}{u_e} u_e' \left(2 + \frac{\delta_1}{\delta_2} - M_e^2\right) = \frac{\tau_w}{\rho_e u_e^2}. \tag{S10.92}$$

The origin of the $-M_e^2$ term follows from the edge-flow relations. For an inviscid edge flow, streamwise momentum and isentropic thermodynamics give

$$p_e' = -\rho_e u_e u_e', \qquad p_e' = a_e^2 \rho_e',$$

where a_e is the edge speed of sound. Equating these expressions and using $M_e = u_e/a_e$ yields

$$\frac{\rho_e'}{\rho_e} = -\frac{u_e u_e'}{a_e^2} = -M_e^2 \frac{u_e'}{u_e}. \tag{2}$$

Schlichting and Gersten state this exact edge relation explicitly as their Eq. (10.100) and note that it follows from the momentum equation together with the condition $T_0 = \text{const}$ at the outer edge of the boundary layer [1]. Thus, Eq. (2) is not an additional relation introduced here, but the outer-flow relation used in Schlichting's derivation of Eq. (10.92). Accordingly, the $-M_e^2$ coefficient in Eq. (S10.92) accounts for the streamwise variation of the edge density.

Multiplying Eq. (S10.92) by $\rho_e u_e^2$ and collecting the δ_1 term separately gives

$$\tau_w = \rho_e u_e^2 \left[\delta_2' + (2 - M_e^2)\frac{\delta_2}{u_e} u_e'\right] + \rho_e u_e \delta_1 u_e'.$$

The bracketed terms form a product derivative. Indeed, the product rule gives

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \delta_2) &= \rho_e u_e^2 \left[\delta_2' + \left(\frac{\rho_e'}{\rho_e} + 2\frac{u_e'}{u_e}\right)\delta_2\right] \\ &= \rho_e u_e^2 \left[\delta_2' + (2 - M_e^2)\frac{\delta_2}{u_e} u_e'\right], \end{aligned}$$

where the second line uses Eq. (2). Substitution into the preceding equation therefore gives

$$\tau_w = \frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \delta_2) + \rho_e u_e \delta_1 u_e'. \tag{3}$$

It remains to identify the first term physically. Define

$$I_m = \int_0^h \rho u \,\mathrm{d}y, \qquad I_p = \int_0^h \rho u^2 \,\mathrm{d}y.$$

Integration of steady mass and streamwise momentum conservation through the boundary layer gives

$$I_m' + \rho_e v_e = 0, \qquad I_p' + \rho_e u_e v_e = -h p_e' - \tau_w. \tag{4}$$

Using the inviscid edge relation $p_e' = -\rho_e u_e u_e'$ to eliminate p_e', and using the first of Eqs. (4) to eliminate v_e, yields

$$\tau_w = \frac{\mathrm{d}}{\mathrm{d}x}(u_e I_m - I_p) + u_e'(\rho_e u_e h - I_m). \tag{5}$$

Now introduce Schlichting's displacement thickness, Eq. (10.95),

$$\delta_1 = \int_0^h \left(1 - \frac{\rho u}{\rho_e u_e}\right) \mathrm{d}y, \tag{6}$$

and note that

$$u_e I_m - I_p = \int_0^h \rho u(u_e - u)\,\mathrm{d}y \equiv J. \tag{7}$$

Equation (5) therefore becomes the physical momentum-deficit balance

$$\tau_w = J' + \rho_e u_e \delta_1 u_e'. \tag{8}$$

Equations (3) and (8) contain identical wall-stress and displacement-thickness terms. Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \delta_2 - J) = 0. \tag{9}$$

The standard normalization that both quantities vanish with the momentum deficit removes the possible x-independent constant. Thus

$$\rho_e u_e^2 \delta_2 = J = \int_0^h \rho u(u_e - u)\,\mathrm{d}y,$$

which, after division by $\rho_e u_e^2$, proves Eq. (1).

For comparison, a literal reading of Schlichting's Eq. (10.96) defines

$$\theta_\mathrm{S} = \int_0^h \frac{\rho u}{\rho_e u_e}\left(1 - \frac{\rho u}{\rho_e u_e}\right)\mathrm{d}y. \tag{S10.96}$$

Multiplying the two candidate definitions by $\rho_e u_e^2$ gives

$$\rho_e u_e^2 \theta_\mathrm{c} = \int_0^h (\rho u u_e - \rho u^2)\,\mathrm{d}y, \tag{10}$$

$$\rho_e u_e^2 \theta_\mathrm{S} = \int_0^h \left(\rho u u_e - \frac{\rho^2 u^2}{\rho_e}\right)\mathrm{d}y. \tag{11}$$

The momentum equation transports ρu^2, as present in Eq. (10), not $\rho^2 u^2/\rho_e$ as present in Eq. (11). Therefore Eq. (S10.96) is not the momentum-deficit coefficient in Eq. (8) for a general variable-density profile. □

3 Consequence and checks

3.1 Exact consequence of using Eq. (S10.96)

Let

$$\Delta\theta \equiv \theta_\mathrm{S} - \theta_\mathrm{c} = \int_0^h \frac{\rho u}{\rho_e u_e}\frac{u}{u_e}\left(1 - \frac{\rho}{\rho_e}\right)\mathrm{d}y. \tag{12}$$

Since $\theta_\mathrm{c} = \theta_\mathrm{S} - \Delta\theta$, the correct balance becomes

$$\tau_w = \frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \theta_\mathrm{S}) + \rho_e u_e \delta_1 u_e' - \frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \Delta\theta). \tag{13}$$

Equivalently, Eq. (10.92) written with θ_S requires the residual

$$\mathcal{R}_S = \frac{1}{\rho_e u_e^2}\frac{\mathrm{d}}{\mathrm{d}x}(\rho_e u_e^2 \Delta\theta) = \Delta\theta' + (2 - M_e^2)\frac{u_e'}{u_e}\Delta\theta. \tag{14}$$

Constant density is a sufficient special case for $\Delta\theta = 0$; it is not an assumption made in the proof.

3.2 Assumptions used

Equation (S10.92) and its original assumptions are accepted. The short derivation of Eq. (8) uses the same standard ingredients: steady two-dimensional thin-boundary-layer conservation, a stationary impermeable wall, a uniform edge state, no streamwise body force, and the inviscid edge momentum relation. Equation (2), already encoded in Eq. (S10.92), is the additional isentropic edge relation needed for its Mach-number form. No constant-density, zero-pressure-gradient, or thermal-wall assumption is introduced.

3.3 Cross-check with Schlichting

Equation (6) is exactly Schlichting's Eq. (10.95), while Eq. (S10.92) is his Eq. (10.92). His Eqs. (10.97) and (10.98) also retain the expected pattern of a mass-flux weight multiplied by a deficit in the transported quantity. Independent standard references print Eq. (1): White and Majdalani, Eq. (7-75), and Anderson, Eq. (6.137) [2, 3]. These checks localize the inconsistency to the second density factor in the printed Eq. (10.96). The argument proves incompatibility of the two printed equations; whether the cause is a typographical error is an editorial question.

References

[1] H. Schlichting and K. Gersten, *Boundary-Layer Theory*, 9th ed., Springer, 2017, Eqs. (10.92)–(10.100).

[2] F. M. White and J. Majdalani, *Viscous Fluid Flow*, 4th ed., McGraw Hill, 2022, Sec. 7-4.2, Eq. (7-75), p. 418.

[3] J. D. Anderson, Jr., *Hypersonic and High-Temperature Gas Dynamics*, 2nd ed., AIAA, 2006, Sec. 6.7.7, Eq. (6.137), p. 333.